

Name:

Task 1. Decide whether the following statements are true or false.

- (i) There exists a 5-regular graph on 21 vertices. **F**
- (ii) There exists a 10-regular graph with 11 vertices. **T**
- (iii) There exists a 111-regular graph with 120 vertices. **T**
- (iv) In every (simple, undirected) graph there are two vertices of the same degree. **T**
- (v) For every graph G there is $\chi(G) \geq \frac{|V(G)|}{\alpha(G)}$. **T**
- (vi) For every graph G there is $\chi(G) \leq \Delta(G) + 1$. **T**
- (vii) For every graph G there is $\chi(G) \geq 1 + \max_{H \subseteq G} \delta(H)$. **T**
- (viii) If tournament is a poset then it has exactly one king. **F**
- (ix) There is no tournament with 2 kings. **F**

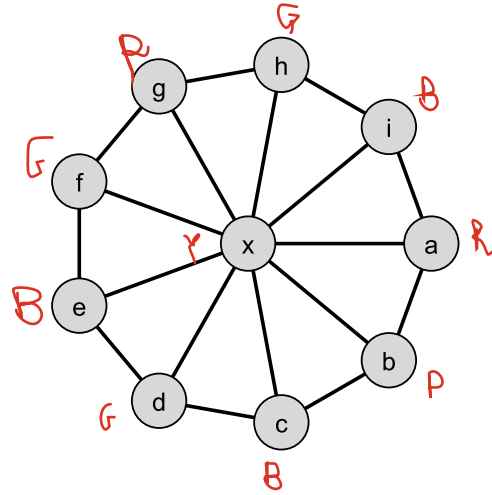
Task 2.

- (i) How many (simple, undirected) graphs exist with the set of vertices $\{1, 2, 3, 4, 5, 6, 7\}$? **2²¹ 2097152 graphs**
- (ii) How many trees exist with the set of vertices $\{1, 2, \dots, n\}$? **n^{n-2} (Cayley's formula)**
- (iii) How many trees on 5 vertices exist up to isomorphism? **3**
- (iv) How many (simple) digraphs on 2 vertices exist up to isomorphism? **4**
- (v) How many perfect matchings has the complete graph on the set of vertices $\{1, 2, 3, 4, 5, 6, 7, 8\}$? **K_8**
 $(2n-1)!! = (2n-1)(2n-3) \dots 1$ $\left. \begin{matrix} 2n=8 \\ n=4 \end{matrix} \right\} 7 \cdot 5 \cdot 3 \cdot 1 = 105$

Task 3. Draw a graph G (and determine the order in which vertices are coloured) that is neither complete graph nor an odd cycle in which greedy colouring requires $\Delta(G) + 1$ colors.

Task 4.

- (i) Colour the graph properly (and optimally). What is its chromatic number?
- (ii) What is its clique number?
- (iii) Draw (mark edges) of a spanning tree given by the BFS algorithm starting from vertex h .
- (iv) Is it a Hamiltonian graph?
- (v) Is it semi-Hamiltonian?



Task 5. Is it possible to make the above graph Eulerian by adding just one vertex (say y) with some incident edges? If so what is $N(y)$? Justify your answer.

Task 6. Draw the Hasse diagram of the ordered set $X = \{1, 2, 3, 4\} \times \{1, 2, 3, 4\}$ with the relation $(a, b) \preceq (c, d) \Leftrightarrow a \leq c \wedge b \leq d$.

- (i) Find the longest chain
- (ii) Find the longest antichain

Task 7. For the graph below, find (mark the edges) a minimum spanning tree. What is its weight? Also, specify the shortest path (write sequence of vertices) from a to h . What is its length?

